

UNIT 2: Data Structures

1. Data Structure

Definition

A data structure is a way of organizing and storing data in memory so that it can be used efficiently.

2. Types of Data Structure

(a) Primitive Data Structure

- Stores single value
- Examples: `int`, `float`, `char`

(b) Non-Primitive Data Structure

- Stores multiple values
 - Examples: Array, Stack, Queue
-

3. Linear and Non-Linear Data Structure

Linear Data Structure

Data elements are arranged sequentially.

Examples:

- Array
- Stack
- Queue

Non-Linear Data Structure

Data elements are not arranged sequentially.

Examples:

- Tree
 - Graph
-

4. Static and Dynamic Data Structure

Static Data Structure

- Memory size is fixed.

- Example: Array

Dynamic Data Structure

- Memory size can change during execution.
 - Example: Linked List
-

5. Array

Definition

An array is a collection of elements of the same data type stored in contiguous memory locations.

6. One-Dimensional Array (1D Array)

Operations on 1D Array

- Traversal
 - Searching
 - Insertion
 - Deletion
 - Sorting
-

7. Searching Techniques

(a) Linear Search

- Searches element one by one
- Works on unsorted array

(b) Binary Search

- Works only on sorted array
 - Faster than linear search
-

8. Sorting Techniques

(a) Selection Sort

- Selects smallest element
- Places it in correct position

(b) Insertion Sort

- Inserts element in its correct position
-

9. Two-Dimensional Array (2D Array)

Definition

A 2D array is an array of arrays arranged in rows and columns.

Operations

- Traversal
 - Sum of two matrices
 - Difference of two matrices
 - Interchanging rows and columns
-

10. Stack

Definition

A stack is a linear data structure which follows **LIFO (Last In First Out)** principle.

11. Operations on Stack

- **PUSH** – Insert element
- **POP** – Delete element

Implementation

- Using Array
 - Using Linked List
-

12. Infix and Postfix Expressions

Infix Expression

Operator is between operands.

Example: $A + B$

Postfix Expression

Operator comes after operands.

Example: $AB+$

13. Conversion of Infix to Postfix

- Use stack
 - Operator precedence is followed
-

14. Evaluation of Postfix Expression

- Scan expression from left to right

- Operands are pushed into stack
 - Operators pop operands and push result
-

15. Queue

Definition

A queue is a linear data structure which follows **FIFO (First In First Out)** principle.

16. Operations on Queue

- **INSERT (Enqueue)**
- **DELETE (Dequeue)**

Implementation

- Circular Array
 - Linked List
-

Very Important Exam Questions

- Define data structure
 - Difference between stack and queue
 - Define linear and non-linear data structure
 - Explain stack with operations
 - Infix to postfix conversion
 - Array searching and sorting
-

Exam Tips

- Draw simple diagrams for Stack and Queue
- Write **LIFO** and **FIFO** in capital letters
- Mention at least 2 points in difference questions